POAMS LECTURES B

Lecture 13: Orbital Motion, 8 pages.

Natural orbital motion, Newton's theory of gravitation, the two-body problem, the role of angular momentum in orbital motion, the role of Newton's gravitational force in orbital motion, conservation of angular momentum.

Lecture 14: The Pope-Osborne Angular Momentum Synthesis, 9 pages.

Newton's theory of gravitation, the Pope-Osborne Angular Momentum Synthesis (POAMS), felt and seen distant actions, the fundamental postulates of the Pope-Osborne Angular Momentum Synthesis (POAMS).

Lecture 15: POAMS and the Two-body Problem, 8 pages.

The fundamental postulates of the Pope-Osborne Angular Momentum Synthesis (POAMS), the POAMS approach to the two-body problem, the POAMS equation of motion.

Lecture 16: POAMS and Circular Orbital Motion, 7 pages.

The POAMS approach to the two-body problem, the POAMS equation of motion, circular orbits, Example One, Example Two.

Lecture 17: More on POAMS and Orbital Motion, 6 pages.

The POAMS approach to the two-body problem, circular orbits, Example Three, the POAMS analogue of Newton's Second Law.

Lecture 18: Angular Momentum and Spin, 7 pages.

Introduction, spin angular momentum, the incorporation of spin into orbital angular momentum in POAMS.

Lecture 19: The Incorporation of Spin into Orbital Motion, 9 pages.

Introduction, the effect of spin in POAMS: 'attraction' and 'repulsion', Example One, Gravity Probe B.

Lecture 20: More on the Effect of Spin on Motion, 8 pages

Introduction, Example Two, the Hayasaka and other experiments.

Lecture 21: POAMS and Spin at the Quantum Level, 10 pages

POAMS at the quantum level, Example Three, 'attraction' and 'repulsion' in the hydrogen atom, summary.

Lecture 22: The Eskridge-Nelson-Schoenfeld Research, 8 pages.

Introduction, a reformulation of POAMS, the spin-coupled force.

Lecture 23: More on the Eskridge-Nelson-Schoenfeld Research, 8 pages.

The spin-coupled force, the Eskridge example, a vector formulation of the spin-coupled force.

Lecture 24: The Eskridge-Nelson-Schoenfeld Experiments, 9 pages.

The spin-coupled force, the orbiting gyroscope experiments, spin-coupled forces acting on atoms with nuclear spin, the Bismuth rotor experiment, the conclusions of ENSTM.